

Real-Time Systems

the British University in
Egypt [BUE]

Lecture 0

Module Outline & .. a brief Intro. to RTS

- Course Details & Course Instructor (Short Bio)
- Tentative Topics & Learning Objectives
- Course Logistics & Grading Policy
- Classroom Code of Conduct
- Be an Effective Learner ..?
- Tentative Weekly Plan

School of
Informatics &
Computer Sciences

SPRING 2026

Module Details

Module Code	25CSSE03H
Module Name	Software Development for Real-Time Systems
Coordinating Unit	School of Informatics & Computer Sciences
Term	Semester 2 [Spring]
Level	Undergraduate – Year 3 (<i>after prep.</i>)
Credit Hours	3.0
Module Leader	Amr S. Ghoneim
Office hours	Sundays – from 1:00 p.m. to 4:00 p.m. ICS Building H – Room 111-E (<i>1st Floor</i>)
Contact	amr.ghoneim@bue.edu.eg
Assigned TAs	TA Zeina Sweilam & TA Alaa Hesham
Grading Scheme	50% - Final Written Exam 50% - Group Practical-Project

Course Staff : Instructor (Short Bio)



Dr. Amr S. Ghoneim

He received his B.Sc. and M.Sc. (by research) degrees in Computer Science from Helwan University (the Faculty of Computers and Artificial Intelligence), Cairo, Egypt. His M.Sc. work on employing AI and machine learning techniques to detect the effects of diabetes on human vision (from retinal images) won 1st Place — Postgraduate [Best MSc/PhD Thesis] — in the MIE “Made-In-Egypt” Competition (organized by the IEEE-Egypt Gold Section) and 1st Place — Postgraduate [Best MSc/PhD Thesis] — in the MIA “Made-In-the-Arab-World” Competition (organized by the Arab League), competing against researchers from 13 MENA countries.

He completed his PhD at the School of Engineering and Information Technology (SEIT), the University of New South Wales at the Australian Defence Force Academy (UNSW@ADFA), Canberra, Australia, where he was a member of the Artificial Life and Adaptive Robotics Laboratory (ALAR). His thesis developed a computational approach for competency awareness in strategic decision-making (for Go players).

Currently, he is an Associate Professor at the Department of Computer Science, Capital University (formerly Helwan University), Cairo, Egypt, where he serves as Head of the Department of Computer Science and Acting Head of the Department of Artificial Intelligence. From 2015 to 2020, he was the Vice-Director of the International Students Office (for Study Programs and Research Communication) at Helwan University. He is also an Adjunct Associate Professor of Computer Science at the Faculty of Informatics and Computer Science (ICS) — the British University in Egypt. His research interests include artificial and computational intelligence, machine learning, ensemble learning, image processing (primarily medical), pattern recognition, statistical analysis, computational psychology, and evolutionary computation.

Tentative Topics & Learning Objectives



.. of this course



Learning Objectives of this Module

Software Engineering for Real-Time System

What you'll learn?

- ✓ ... the **basic concepts of real-time systems** (*including concurrency, distributed systems, embedded systems*),
- ✓ ... the **special requirements** that they place on the **design process**, ... and
- ✓ ... the **software tools** used for their design and implementation.

Topics Covered [Tentative / Indicative]

The main contents of the course are:

Advanced Applied Software Engineering, Modelling, & Development, while focusing on Real-time Software Engineering (Software Modelling & Development for Real-time Systems).

Topics Covered [Tentative / Indicative]

The main RTS concepts covered in the course are :

- **Fundamentals of Real-time Systems** (*Response Time, Real-Time System vs. Embedded System, Soft, Hard, & Firm “Real-Time”, CPU Utilization, etc.*).
- **UML for Concurrent Applications, Real-time, & Distributed Applications**
 - COMET (*C*oncurrent *O*bject *M*odelling & architectural design *mETHod*).
 - An Intro. to UML State-Chart Diagrams, and Hierarchical State-Charts.
 - UML Extension Mechanisms.
 - Modelling Application Classes and Objects (*Entity & Physical Classes, Boundary Classes Vs. External Classes, Control Classes & Objects*).
 - UML Dynamic Modelling: Concurrent Collaboration Diagrams.
- **Concurrent Processing**
 - The Mutual Exclusion Problem (*Multiple Readers & Writers Problem*).
 - Task Synchronization Problem (*Concurrent Robots Systems*).
 - The Producer/Consumer Problem.
- **Designing Concurrent & Real-Time Software Architectures** (*Software Architectural Patterns, Control Patterns for Real-time Software Architectures*).
- **Real-time Design Patterns**



Course Logistics

- Announcements & Course Materials
- Additional Resources (per section)
- Assessment Summary (Grading Policy)
- Exercises .. What have we learned?
- Be an Effective Learner ..?
- Classroom Code of Conduct
- Academic Integrity & Plagiarism

Announcements & Course Materials [1]

Downloads, Homework, & Announcements:

All course material (*lecture notes “slides”, announcements, assignments, any supplemental notes or documentation*), will be made available “posted” online on weekly basis, on the eLearning: <https://learn1.bue.edu.eg/>

Announcements & Course Materials [2]

Video Lessons:

Only Some relevant recorded lectures can be found on the following:

Amr S. Ghoneim YouTube Channel : <https://www.youtube.com/AmrSGhoneim/>

Software Development for Real-Time Systems YouTube Playlist :

<https://www.youtube.com/playlist?list=PLsnvpvHuTUbC-Yba-gGn9UjCZB7-abGWS>

Software Engineering 1 YouTube Playlist :

https://www.youtube.com/playlist?list=PLsnvpvHuTUbC-yJkvcf-Stp_kLwfesnn-

Announcements & Course Materials [3]

Reading List:

For Real-Time (Concurrent) Systems Analysis, Design, & Modelling:

- **Jim Cooling, (2019). The Complete Edition – Software Engineering for Real-Time Systems - A Software Engineering Perspective Toward Designing Real-Time Systems, Packt Publishing.**
- **Hassan Gomaa, (2000). Designing Concurrent, Distributed and Real-time Applications with UML, Addison Wesley, ISBN: 978-0201657937.**
- **P. A. Laplante, and S. J. Ovaska, (2012). Real-Time Systems Design and Analysis: Tools for the Practitioner, Fourth Edition, IEEE press.**
- **Bruce Powel Douglass, (2002). Real-Time Design Patterns: Robust Scalable Architecture for Real-Time Systems, Addison Wesley.**
- **Cooling, J., (2002). Software Engineering for Real-time Systems, Addison Wesley, ISBN: 978-0201596205.**
- **Gomaa, H., (2011). Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures, Cambridge University Press.**
- **EsperTech: Event Series Intelligence;**
<http://www.espertech.com/esper/tutorials.php>

Announcements & Course Materials [4]

Reading List:

For OO Design Principles, Analysis & Design, UML Modelling:

- Bernd Bruegge, and Allen H. Dutoit , (2009). **Object-Oriented Software Engineering: Using UML, Patterns and Java**, 3rd Edition, Prentice Hall.
- Hassan Gomaa , (2011). **Software Modeling and Design: UML, Use Cases, Patterns, and Software Architectures**. Cambridge University Press.
- Martina Seidl, et al., (2015). **UML@ Classroom**. Springer.
- Martin Fowler , (2004). **UML Distilled: a Brief Guide to the Standard Object Modeling Language**. Addison-Wesley Professional.
- B. Unhelkar , (2017). **Software Engineering with UML**. Auerbach Publications.
- Matt Weisfeld , (2019). **The Object-Oriented Thought Process**. 5th Edition.

Announcements & Course Materials [5]

Reading List:

For Software Engineering:

- Ian Sommerville, (2020). **Engineering Software Products – An Introduction to Modern Software Engineering**. *Pearson*.
- Ian Sommerville, (2016). **Software Engineering 10th Edition**. *International Computer Science*.
- G. O'Regan, (2017). **Concise Guide to Software Engineering**. *Springer, Cham*.

For the Design Patterns:

- J. Vlissides, R. Helm, R. Johnson, and E. Gamma , (1995). **Design patterns: Elements of reusable object-oriented software**. *Reading: Addison-Wesley*.
- E. Freeman, E. Robson, B. Bates, K. Sierra, (2004). **Head first design patterns**. *O'Reilly Media Inc*.

Assessment Summary (Grading Policy)

Assessment Task 1: Final Unseen Written Exam - 50%

Goal: Assessment of what you have learned from the lectures, tutorials, and the term project.

Product: Unseen Written Exam.

Format: The final two hours exam will be conducted at the end of the semester at a time and place to be announced.

Generic skill assessed:

- Applying technologies.
- Knowledge and understanding.
- Problem solving.

Assessment Summary (Grading Policy)

Assessment Task 2: Group Practical Project - 50%

Goal: You will demonstrate your cognitive, design, modelling and technical skills in real-time systems design and development.

Product: Two deliverables are expected:

1. Analysis and Design document.
2. Running prototype using the technologies learnt.

Format: A design and programming project (4 to 5 students), to design and develop a real time application, with one report (design) and one prototype. The students will also be discussed individually.

Generic skill assessed:

- Applying technologies.
- Problem solving.

Assessment Summary (Grading Policy)

Assessment Task 2: Group Practical Project - 50%

Indicative Marking Scheme: 100 Marks distributed as follows:

55 Marks for the analysis and design report:

- Extended Use Case diagram. *10 marks*
- Stimuli/Response Identification. *08 marks*
- Detailed Class & Object diagrams. *12 marks*
- State-Machine diagrams. *15 marks*
- Interaction diagrams (Sequence and/or Collaboration). *10 marks*

45 Marks for the code/discussion (based on the following criteria):

- 5 weekly discussions (individually graded). *05 marks*
- Final individual discussion. *10 marks*
- Requirements are fulfilled. *05 marks*
- Correctly mapping design models into executable code. *05 marks*
- Running correctly and Detailed Testing. *05 marks*
- Correct multithreading implementation and synchronization. *10 marks*
- Using Authentication & Updating UI in real-time. *05 marks*

Academic Integrity & Plagiarism

- You can discuss ideas and methodology for the homework or the project with other students in the course, but ***you must write your solutions completely independently.***
- We will be code-checking to assess similar submissions or submissions that use code from other sources.
- You ***may*** use snippets of code from other sources, as long as you cite these properly (put a comment above and below whatever portion of code is copied); but **be reasonable.**
- Please note that, in case there are any **concerns about AI-generated submission content**, you will be invited for an opportunity to verify and defend your work.

Additional Resources

- At the end of some sections, there will be a short list of additional resources.
- The list will include articles, YouTube videos, online courses, books, .. etc. (in both Arabic and English).
- These resources will provide more details on some of the topics discussed during the lectures.

Exercises

What have we learned?

At the end of some sections, there will be a set review activities (exercises), that would help you see the progress you've made in this lesson.

The exercises would help you:

- Highlight what you have learned today & emphasize key information.
- Correct misunderstandings.
- Summarize, review, & demonstrate your understanding of major points.
- Transfer ideas to new situations.

Classroom Code of Conduct

Be punctual & prepared to study:

- Attend all classes including online and face-to-face classes.
- Arrive to all classes on time.
- Keep all handouts & work in a folder and make sure they are well organised.
- Prepare yourself for your classes.

Participate in the classroom:

- Participate in class.
- Respect yourself, your teachers, and your classmates.
- Turn mobile phones off or put them on silent mode before entering the classroom.
- Adopt a professional attitude - no eating, side-talk, etc.

Study independently:

- Study outside the classroom as part of your student effort (*about 6 to 8 hours per week*).
- Complete all homework assignments and hand them in on time.
- Use the University/Faculty facilities (*e.g., the library*) and online resources.

Respect University regulations:

- Follow (University, Faculty, Department, and Programme) regulations.
- Know the due date of all assessments, submit them by that date & in the required format.
- Understand what you need to successfully progress.
- Maintain academic honesty (*academic integrity & plagiarism*).

Be an Effective Learner ..

- Have the desire to seek knowledge and acquire new skills (be inquisitive / curious).
- Ask questions.
- Be an avid (keen & passionate) reader.
- Be an attentive / focused listener.
- Find your preferred learning style .. & .. Learn in multiple ways.
- Do NOT memorize.
- Embrace Discomfort.
- Practice, practice, practice (you must gain practical experience).
- Teach what you've learned to another person.
- Use testing to boost / improve learning.
- Avoid multitasking.
- Make use of Memory Improvement Basics.
- Draw up a schedule.
- Examine your lifestyle.
- Create a study station.

An Introduction to **Real-time Systems**

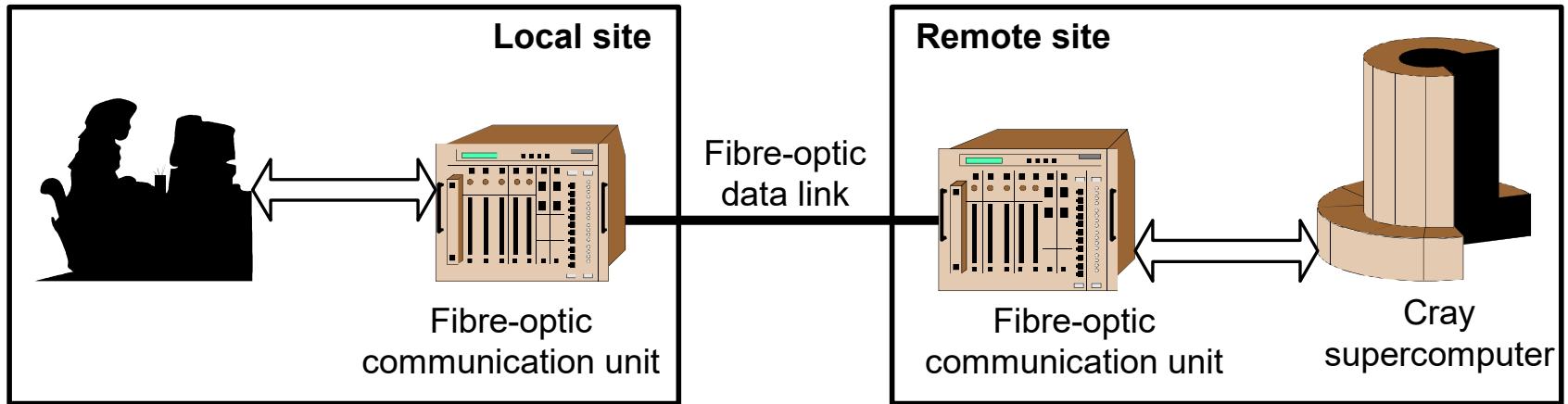
Introduction to Real-time Systems .. Objectives

- Outline the **differences** between **general-purpose** computer applications and **real-time** systems.
- Give an **overview** of practical **real-time** system **structures**.
- Describe some **key environmental** and **performance** requirements of **real-time** systems.
- Technology **trends** enabling the **rapid evolution** of real-time systems.

Categorizing Computer Systems

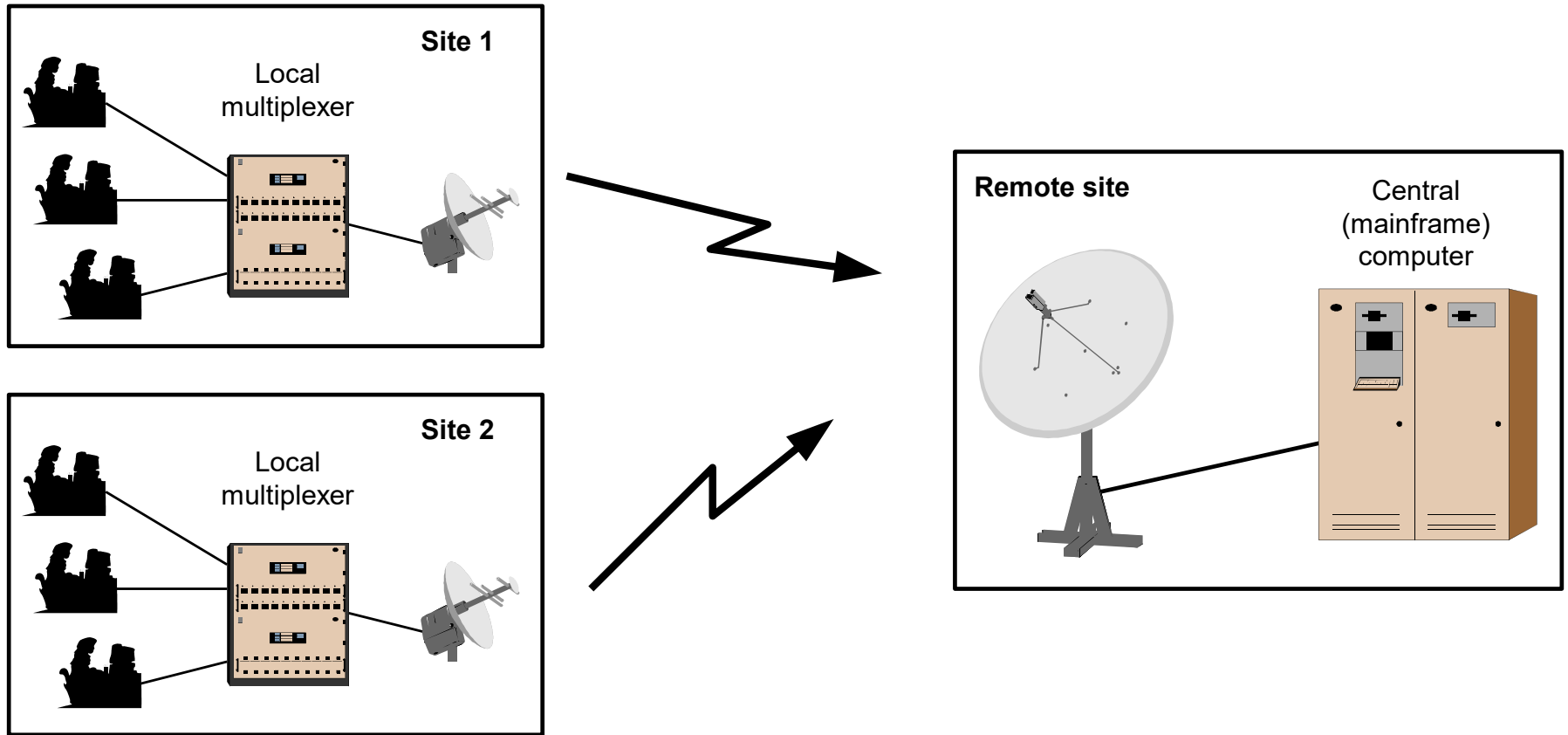
- Multiple categorizations do exist (*based on size, computational power, technology/generation, intended usage, etc.*).
- If we choose the **speed of response** as the main criterion, **three** general groups emerge:
 - **Batch**: I don't mind when the computer results arrive. The time taken may be hours or even days in such systems.
 - **Interactive on-line**: I would like the results with a *fairly short time*, typically a *few seconds*.
 - **Real-time**: I need the results within definite time scales, otherwise the system just *won't* work *properly*.

An Example of a Modern Batch System



- Methods like this are used where **computing resources** are **scarce** and/or **expensive**, as it is very efficient technique
- The user usually **pre-processes** all programs and information (**storing** on a **local** system computer)
- At a convenient time, this **job** is **passed** over the data link to a **remote site**
- When all jobs are **finished**, the **results** are **transmitted** back the **originating** site.

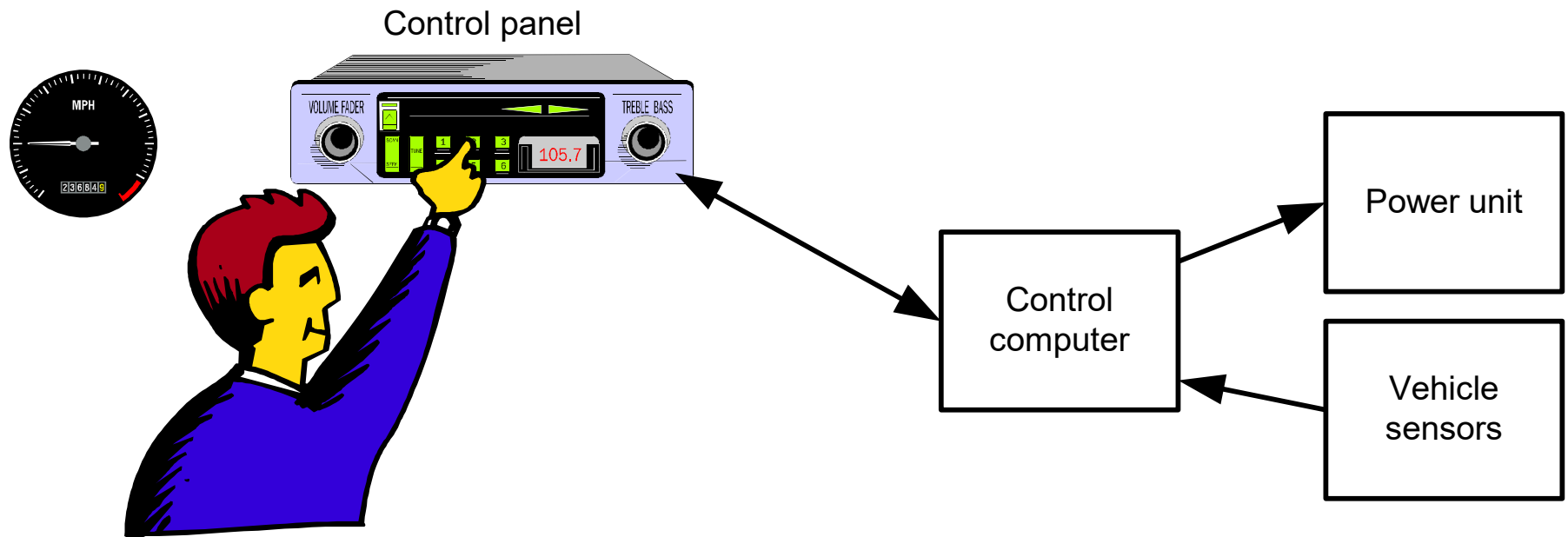
A Typical Interactive On-line Computer System



A Typical Interactive On-line Computer System

- Interactive online computer systems are **widely** used in **banking, travel bookings, and mail-order** systems.
- All **transactions** are handled by?
 - the **central computer** in a time-sliced fashion.
- **Routing and access control** are the responsibility of?
 - the **front-end** processors and local multiplexers.
- Typical model of the **internet**.
- **Response time** depends on?
 - the **amount of activity**; all systems **slow-down** as **load** builds up, sometimes **seizing-up** at **peak times**.

An Example of a Real-time Computer System: Auto-Cruise Control System



- For time critical applications, such as an auto-cruise control system, this **response time** is **unacceptable**.

An Example of a Real-time Computer System: Auto-Cruise Control System

- The driver dials in the desired cruising speed.
- The cruise control computer notes this and compares it with the actual vehicle speed.
- If there is a difference, correcting signals are sent to the power unit.
- The vehicle will either speed up or slow down.

So .. What is a Real-time Computer System?

- A real-time system is a system that **must satisfy explicit** (bounded) **response-time** constraints, or **risk severe** consequences, including **failure**.
- A real-time system is one whose **logical correctness** is based on **both** the **correctness** of the **outputs** and their **timeliness**.
- A real-time system is any **information processing** activity or **system** which must **respond** to **externally** generated **input stimuli** within a **finite** and **specified** period.



**Welcome to
CSSE03H**

**Up Next ..
Lecture 1**

Software Development for Real-Time Systems

Thank you!